

Temporal 3D Convolutional Neural Network for Predicting Radiation Pneumonitis within 90 days

Brændstrup, Victor Holler; Garner, Natasja Toftgaard; Kristiansen, Signe Wittrup; Lau, Signe Meiners; Schaldemose, Rasmus Blegvad

Biomedical Engineering and Informatics MSC, Department of Health Science and Technology, Aalborg University

Abstract

Background & Aim: Radiation-induced pneumonitis is a lung inflammatory condition, caused by exposure to ionizing radiation, often from undergoing cancer treatment. RP develops 1–3 months post radiation therapy and ranges from asymptomatic to life-threatening. While studies have explored machine learning for RP prediction, current models lack clinical utility. This study aims to develop a CNN model to predict whether a patient will develop RP within 90 days, based on radiological findings between two time points.

Methods & materials: CT scans from 179 patients with non-small cell lung cancer were examined to identify features in baseline and follow-up scans. Each baseline scan and its corresponding follow-up scan were rigidly registered and lung segmented. Features were extracted using a pre-trained ResNet-50. Baseline and follow-up features were concatenated and fed into a custom CNN classifier to predict radiation pneumonitis within 90 days. The performance of the CNN was evaluated using F1 score, precision, recall, and a confusion matrix.

Results: Three versions of the CNN were trained using different sampling and weighting strategies. WR used balanced sampling (F1=0.552). WRR25 applied 75% of original sample weights (F1=0.354). WRR50WL reduced minority class weight to 50% (F1=0.632).

Conclusion: The performance of the model demonstrated the potential of a deep learning model to predict RP within 90 days. However, current performance is insufficient for clinical use.

Introduction

Radiation-induced pneumonitis (RP) is lung inflammation caused by ionizing radiation and is therefore frequently seen in patients undergoing treatment for lung cancer.[1] Lung cancer is the most commonly diagnosed cancer worldwide, accounting for 12.4% of all cases, and remains the leading cause of cancer-related deaths.[2] RP typically occurs 1–3 months after radiation therapy (RT) and can be present in various ways, ranging from subclinical findings to life-threatening disease requiring hospitalization.[3] Furthermore, the incidence of RP is 15–40%, and the risk of mortality associated with RP is

estimated to be under 2%.[4] The most common symptoms of RP are fatigue, dry non-productive cough, mild to severe dyspnea, fever, and sometimes pleural pain. If damaged lung tissue is advanced enough, RP could lead to respiratory failure.[3] To rapidly provide the most optimal treatment, an early diagnose of RP is important[5], but no reliable prediction methods currently exists.[6]

The largest risk factors for evolving RP after RT are the total dose of gray and the amount of gray per fraction. The usual dose used for most cancers exceeds 50 Gy, which is above the threshold associated with a higher risk of developing RP.[3] Furthermore, if more than 50% of the lung volume is irradiated, there is a greater prevalence of RP.[3]

Due to these risk factors, prevention becomes difficult, and early diagnosis needs further investigation.

The diagnosis of RP can be challenging, as its clinical symptoms and radiological findings are often similar to those of other lung diseases, and it is therefore often diagnosed by elimination of differential diagnoses.[3] In spite of this, a recent study by Szejniuk et al.[7] showed a strong correlation between the number of lung segments affected by radiological computed tomography (CT) findings and whether a patient would develop RP. These findings include ground-glass opacities, consolidations, atelectasis, lung volume reduction, and fibrosis. Additionally, the study showed that patients with more affected lung segments had a higher chance of developing RP earlier, suggesting that radiological features could be a useful tool for detecting RP early on. Predicting RP in advance may allow for earlier intervention, potentially improving patient outcomes. A potential way to achieve this is by implementing deep learning methodologies such as convolutional neural networks (CNN) as they are designed to learn meaningful patterns in image data such as CT scans.[8]

In a study by Lee et al. [9], a deep learning model called MergeNet was developed to predict the risk of RP in patients after RT. The study combines pretreatment chest CT images with clinical factors to predict the risk of RP where the CT images were processed in a convolutional neural network (CNN) and the clinical factors in a fully connected neural network. The two structures were merged together to predict the risk of RP. MergeNet achieved an AUC of 0.689 suggesting that the combination of imaging data and clinical factors may contribute to risk assessment.

Another study by Kawahara et al. [10] predicts RP after stereotactic body radiotherapy (SBRT) by extracting features from 3D dose distributions in the normal lung using a pretrained 2D CNN (ResNet-50). Combining these features with dosiomics features, results in a model (AUC = 0.84), which is significantly improved compared to the model without dosiomics features (AUC = 0.58).

Overall, deep learning models analyzing CT scans and dose data show strong potential for improving RP risk prediction. These findings collectively highlight both the clin-

ical need and the technical feasibility of developing deep learning models to support early and accurate prediction of RP.

Given the importance of early detection, there is considerable interest in predicting whether symptomatic RP will develop. A study by Li et al.[11] showed promising results of using medical imaging for assessing disease progression by using machine learning through a convolutional Siamese neural network. It was applied to evaluate disease severity in both retinopathy of prematurity and knee osteoarthritis by comparing images from different time points for each patient.

Additionally, results from CNN change detection models in non-medical fields show promising results in their ability to identify changes by comparing images taken at different time points. Both approaches use aligned image pairs to predict pixel-level change maps. Daudt et al. employ an encoder-decoder Fully Convolutional Network design [12], while Varghese et al.'s [13] ChangeNet relies on a Siamese ResNet backbone .

Based on these findings, this study aims to develop a CNN model to predict whether a patient will develop RP within 90 days, based on radiological findings between two time points.

Methods

Data

Data from 179 patients diagnosed with non-small cell lung cancer were retrospectively analyzed. The data was approved by the North Denmark Region Committee on Health Research Ethics (reg. no N-20120029) and registered to the Danish Data Protection Agency (2008-58-0028).[7] All patients received RT at the Department of Oncology, Aalborg University Hospital, between February 2014 and January 2021. The total radiation dose ranged from 60 to 66 Gy, administered in 30 to 33 fractions, respectively. For patients diagnosed with RP, the diagnosis date was recorded if RP was graded 2 or higher according to the Common Terminology Criteria for Adverse Events

scale (CTCAE). All scans were acquired with patients in the supine position, either head-first or feet-first, however baseline and follow-up scans were acquired using different CT scanners and protocols. For the baseline scans, patients were instructed to lie with their arms above their head and to breathe normally. For the follow-up scans, no specific instructions were given regarding patient position, but patients were instructed to hold their breath during the scan. The slice thickness across all CT scans ranged from 1.25 mm to 2.5 mm. All CT scans were anonymized and manually restricted to the thoracic region of the patient.

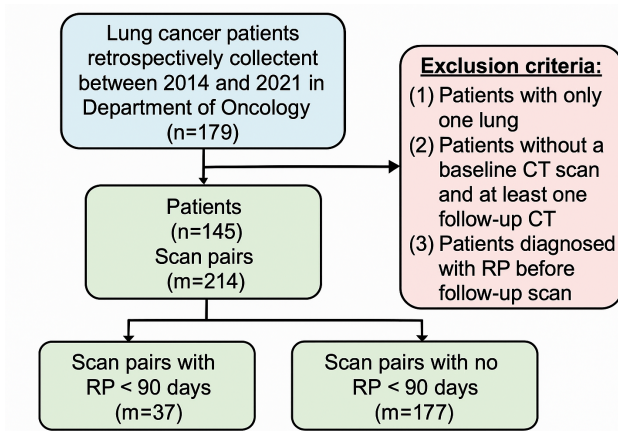


Figure 1: Flow diagram over included patients and scan pairs, with corresponding exclusion criteria where n is the number of patients and m is the number of scan pairs.

Patients were included if they had a baseline thoracic CT scan prior to RT and a follow-up CT scan either one month, four months, or both after completing RT. Patients were excluded if they (1) had only one lung, (2) did not have a baseline CT scan and at least one follow-up CT scan, or (3) were diagnosed with RP prior to the follow-up scan as shown in figure 1.

The CT scans were grouped into pairs, each consisting of a baseline scan and a corresponding follow-up scan. If a patient had two follow-up scans, two separate pairs were created using the same baseline scan paired with each follow-up scan. After exclusions, a total of 145 patients were included, resulting in 214 pairs of CT scans as shown in figure 1. Each scan pair was labeled as 1 if the patient was

diagnosed with RP within 90 days of the follow-up scan ($n = 37$), or 0 otherwise ($n = 177$).

Pre-processing

Resampling

All volumes were resampled twice during pre-processing. First, to ensure uniform voxel dimensions across all volumes, they were resampled to isometric voxel dimensions of $1 \times 1 \times 1 \text{ mm}^3$. This was achieved by extracting spatial metadata from the DICOM headers, such as pixel spacing, z-spacing, origin, and orientation. Linear interpolation was used to estimate new intensity values during resampling. Second, to ensure optimal performance during the feature extraction phase, scans were z-score normalized. Additionally, volumes were padded using their respective intercept slope values, and then resampled to a size of $224 \times 224 \times 224$, as this is the expected input dimension of the 3D ResNet-50 model.[14, 15] Although, the result of this resampling lead to a different spacing than $1 \times 1 \times 1 \text{ mm}^3$, the spacing across all scans remained the same due to the identical cubic size of all scans.

Lung segmentation

Lung segmentation was performed to ensure the CNN focused solely on lung tissue, excluding irrelevant surrounding organs and structures. The segmentation was done using an lung segmentation model U-net(R231) by Hofmanninger et al.[16]. The model is trained on various CT-scans of the chest including lungs with a broad range of pathologies and is able to include high-density abnormalities such as tumors and pleural effusions. The results of the Hofmanninger study showed a Dice Similarity Coefficient (DSC) of 0.98 ± 0.03 . [16] All CT scans were pre-processed by converting raw DICOM values to Hounsfield Units (HU), as the U-net(R231) model requires input images with pixel intensities expressed in HU.

Image registration

To enable spatial comparison between baseline and follow-up scans, each follow-up scan was rigidly aligned to its corresponding baseline scan using anatomical landmarks. To avoid misalignment because of inspiration differences across scans, stable landmarks were chosen. To achieve this, the open-source tool TotalSegmentator [17, 8] was used to extract vertebral landmarks for all scans at T4, T8, and T10. 3D coordinates in voxel-space were found by calculating the center of mass for each segmentation mask for all the chosen landmarks. Based on these coordinates, a rigid transformation matrix containing a translation vector and a rotation matrix was computed. This matrix was then applied to the follow-up CT volumes

Augmentation

To increase the datasets variability, data augmentation was applied in a pairwise manner, meaning that identical transformations were applied to each baseline–follow-up scan pair to preserve anatomical consistency. All scan pairs were rotated by a randomly sampled angle within ± 10 degrees. Additionally, scans were flipped along the coronal plane. Finally, a combination of both rotation and flipping was applied. This process resulted in three augmented versions of each original pair: one rotated pair, one flipped pair, and one pair that was both rotated and flipped.

To enhance diversity and robustness, further data augmentation was applied to some scans probabilistically. These augmentations consisted of Gaussian noise and random sharpness adjustments. The augmentations were applied before volumes were passed through the ResNet-50 model.

RP Classification model

The overall model architecture was designed as illustrated in figure 2. The first part of the model is a pre-trained ResNet-50 from MONAI, used to extract features from both baseline and follow-up CT scans. The feature maps from each scan pair were extracted at three different lay-

ers and concatenated into a single tensor, preserving both spatial and temporal information. This concatenated feature tensor was used as an input to a custom CNN model, which was trained to classify whether the patient would develop RP within 90 days after a given follow-up scan.

ResNet-50

For feature extraction, the 3D ResNet-50 model from MONAI was used, which is pre-trained on a large and diverse set of medical imaging data, spanning various modalities and anatomical regions.[18] The model was used solely as a feature extractor and weights therefore remained non-trainable. Consequently, the final fully connected classification layer was removed from the architecture. Both baseline and follow-up 3D scans were used as input for the model. Feature maps were extracted from three layers of the network: layer2, layer3, and layer4, to capture hierarchical features at increasing levels of detail. These layers correspond to progressively deeper levels of the network.[13] The dimensions and number of channels for the feature maps at each layer are as follows:

Layer 2: (1, 512, 56, 56, 56)

Layer 3: (1, 1024, 28, 28, 28)

Layer 4: (1, 2048, 14, 14, 14)

Concatenation

The feature maps for each scan pair were combined to integrate information across the three ResNet layers.[13] Since the feature maps from different layers had varying spatial dimensions, all feature maps were resampled using trilinear interpolation to match the spatial resolution of the layer3 feature maps. This ensured consistent spatial dimensions for subsequent stacking.

For each layer, the baseline and corresponding follow-up feature maps were stacked along the channel dimension, effectively capturing temporal changes between the two time points. The stacked feature maps from the three layers were further stacked along the channel dimension, combining spatial and temporal information into a single tensor with dimension (1, 7168, 28, 28, 28). The process

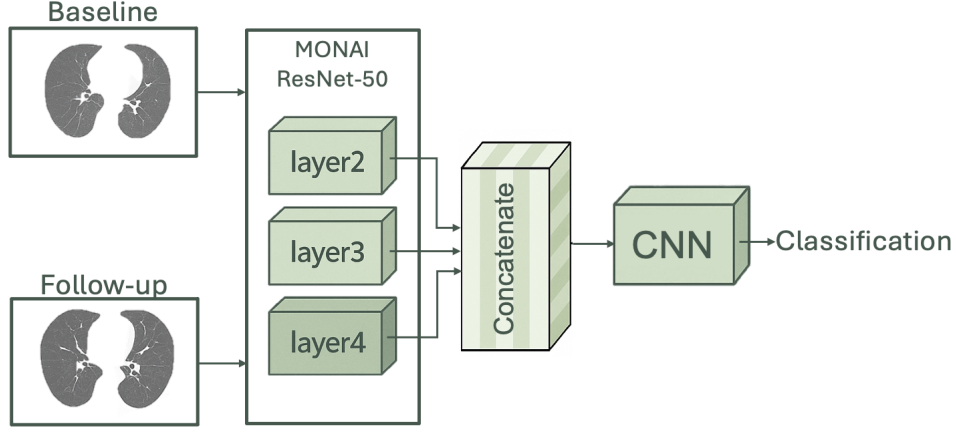


Figure 2: Overview of the model as a whole. The model takes baseline and follow-up CT-scans (1 or 4 months) as input. It extracts feature maps from each scan, concatenates them and subsequently uses a CNN to classify whether the patient will develop RP within 90 days.

was repeated for all scan pairs to enhance model robustness.

CNN

To classify the concatenated feature maps, a custom 3D CNN with five blocks was implemented which is illustrated in figure 3. The architecture of the model is based upon the VGG-Net architecture, as it showed promising results in image classification.[19] Given the limited dataset, it was chosen to implement a reduced depth network architecture. ReLU activation and batch normalization were applied after each convolutional and fully connected layer, except for the final output layer.

Block 1: Reduced the number of channels from 7168 to 2368 using two convolutional layers with a kernel size of 3, and reducing the spatial dimensions to 24x24x24.

Block 2: Reduced the number of channels to 768 using two convolutional layers with a kernel size of 3, and reducing the spatial dimensions to 12x12x12 using a max-pooling layer with a stride of 2.

Block 3: Reduced the number of channels to 256 using three convolutional layers and a max pooling layer with a kernel size and stride of 2, reducing the spatial dimensions to 6x6x6.

Block 4: Preserved the spatial resolution at 6x6x6 and

reduced the number of channels to 64 through three 3D convolutional layers with padding.

Flatten and fully connected: Flattened the output tensor into a 1D vector with a length of 13824, followed by three fully connected layers with output sizes of 5488, 512, and 2, respectively, to generate the final class prediction.

Hyperparameter optimization

Loss function and optimizer

The loss function utilized for the model is the cross entropy loss function from the PyTorch library which frequently is used for binary classification. The loss function serves as a feedback tool for the model, and measures the difference between actual output and expected output. The optimizer used is the Adaptive Moment Estimation (Adam) optimizer to minimize loss, by finding errors based on observation of input and weight combination.[20]

Splitting

Prior to model training, the dataset was split into stratified training and test datasets, by an approximate 80/20 ratio. Subsequently, the training set was further split and stratified into a training and validation dataset by the same

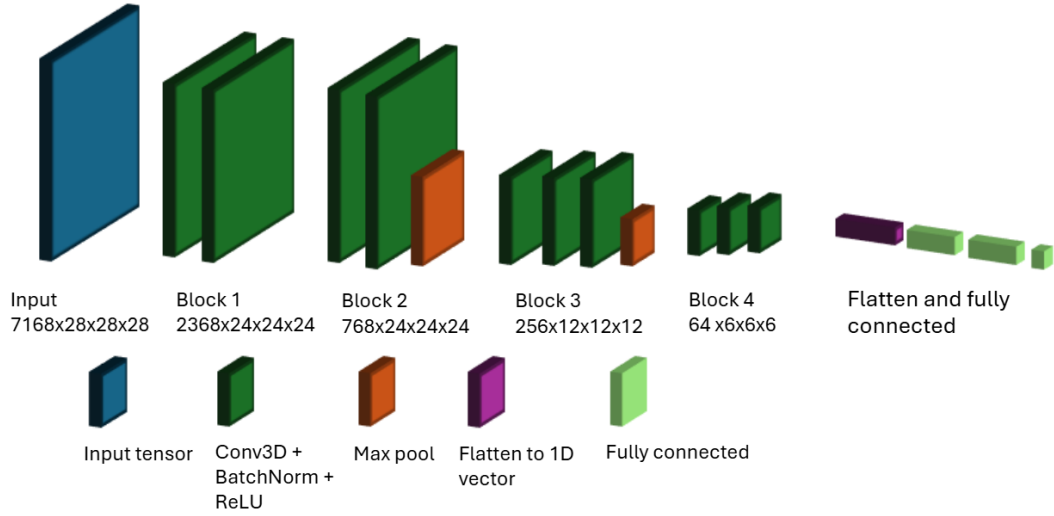


Figure 3: Overview of a custom 3D CNN model with with four convolutional and one fully connected block. The model consists of four similarly structured convolutional blocks, however block 2 and block 3 also includes max pooling. Lastly, a fully connected classifier block produces the output. Below each block, the input dimension is described.

ratio. Stratifying was done by using patient ID and target labels, to reduce risk of data leakage and ensure the distribution of RP in the training and validation datasets matches that of the entire dataset.

Training

The model was trained for 100 epochs using a batch size of 3. The Adam optimizer had a learning rate of $1 * 10^{-4}$, same for the weight decay parameter. To identify appropriate parameters, an iterative manual tuning process was used. This involved adjusting the values based on model performance after each training session. Three different sampling and weighting strategies were explored. Due to significant class imbalance in the dataset, all three versions implemented sample weighting. Each input is assigned a weight inversely proportional to its class frequency, ensuring the minor class samples were more likely to be drawn. This approach aims to help prevent bias toward the majority class.

Weighted Resampling (WR): samples from both classes were drawn with equal probability, resulting in balanced batches consisting of 1:1 ratio of class 0 and 1 instances.

Weighted Resampling Reduced 25% (WRR25): used weighted resampling reduced by 25% corresponding to us-

ing 75% of the original number of samples where the sampling probabilities defined by the weight remain the same.

Weighted Resampling Reduced 50% + weighted loss function (WRR50WL): sample weights for the minority class was reduced to 50% of the full sample weighting. Additionally, the cross-entropy loss function was configured with explicit class weights to amplify the contribution of minority class errors to the training loss, thereby preserving learning sensitivity despite less aggressive sampling.

Model evaluation

To optimize the model and facilitate comparison with other studies, several metrics were calculated. These metrics mainly focused on evaluating the performance of binary classification. The calculated metrics used in this study include precision, recall, and F1 score. This is further supported by a confusion matrix.

Results

The performance for training and validation of the three versions is presented in figure 4, where the performance metrics of precision and recall throughout 100 epochs are visualized.

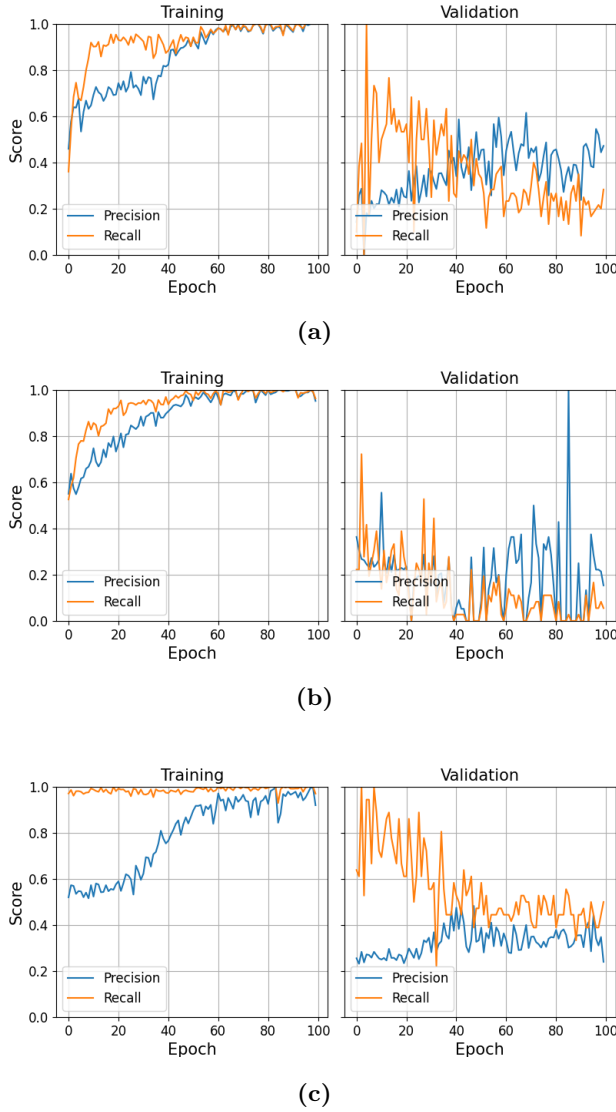


Figure 4: Plots of training and validation metrics for the three model versions where the models are a) WR, b) WRR25, c) WRR50WL.

During training of the model versions, checkpoints corresponding to the highest F1 score in the validation was saved and subsequently used for testing. The optimal checkpoints for each model version were epoch 41 for WR, epoch 2 for WRR25, and epoch 40 for WRR50WL. The tests was performed on the optimal checkpoint for each model version.

In figure 5, the confusion matrices for the prediction on the test dataset of the three versions are visualized.

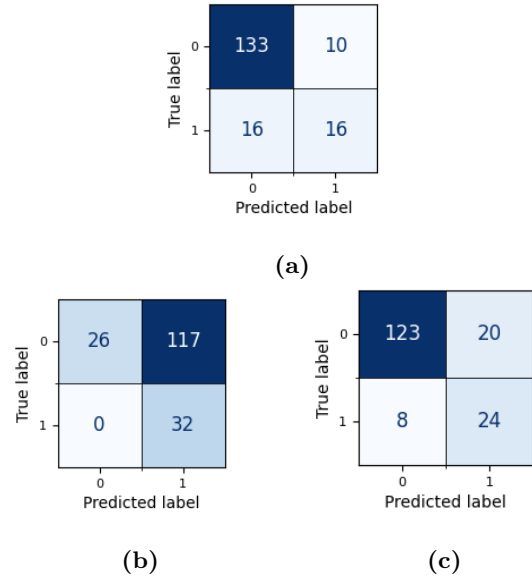


Figure 5: Confusion matrices for test set classification performance of the three trained model versions. The model versions are: a) WR, b) WRR25 and c) WRR50WL

The performance metrics of the three versions on the test data are presented in table 1.

CNN 5 blocks	F1 score	Precision	Recall
WR	0.552	0.615	0.500
WRR25	0.354	0.215	1.000
WRR50WL	0.632	0.545	0.750

Table 1: Performance results for test of three model versions, the results includes F1 score, precision and recall.

Discussion

In this study, a CNN model was developed to predict whether a patient would develop RP within in 90 days after a given CT scan. The model was based on radiological changes observed between two time points: a baseline and a follow-up scan. Different model versions were tested, where model WRR50WL achieved the best performance obtaining an F1 score of 0.632, a precision of 0.545 and a recall of 0.750 which is seen in table 1. The recall score suggests that the model is relatively effective in identifying true RP cases. This is illustrated in figure 5c where the WRR50WL model correctly identifies 24/32 actual RP

cases. However, the precision score reflects a considerable rate of false positives which is seen in figure 5c, where the model is able to correctly identify 24 true RP cases out of 44 cases predicting RP. While it may be acceptable in a clinical context to have a higher recall to prioritize early detection of RP, the trade-off between recall and precision indicates a need for further refinement of the model.

Similar studies using CNNs to predict RP from CT scans reported comparable performances. Lee et al.[9] achieved an F1 score of 0.532, precision of 0.527, and recall of 0.502 using only CT scans as input. Kawahara et al.[10] reported an accuracy of 0.60, specificity of 0.817 and a recall of 0.364. These results, when compared to those in this study, suggests that establishing a reliable method for predicting RP remains a challenge. However, both studies demonstrate that incorporating dosimetric information alongside CT scans can improve model performance.[9, 10] In particular, Kawahara et al.[10] developed a model that only included lung tissue receiving radiation doses exceeding 20 Gy. Their model achieved an accuracy of 0.757, specificity of 0.8 and recall of 0.709, suggesting that dosimetric information may be relevant in order to improve the predictive performance of RP.

The dataset was limited to 214 scan pairs after exclusions, which generally is considered a small size for training deep learning models. To address this, various augmentation techniques were applied, resulting in an expanded dataset of 856 scan pairs. This inflation of the dataset was intended to enhance the model’s ability to learn robust and generalizable features. Since deep learning models typically require large datasets to extract relevant patterns effectively, dataset size is a critical factor in achieving successful model performance.[18] A substantially larger dataset would likely have improved the performance of the model, but it warrants further investigation.

The labeling scheme was designed to be able to predict RP within 90 days of a given scan. Similar studies by Lee et al. and Kawahara et al. [9, 10] did not have a time constraint on the prediction of RP, which is likely to increase the difficulty of the prediction task in this study. However, this design choice is based on the clinical importance

of being able to detect RP early in the disease course.[5] Furthermore, the time constraint is physiologically justified, as Szejniuk et al.[7] reported a strong correlation between the extent of lung tissue damage after RT and the likelihood of developing RP symptoms within 8 weeks.

The class distribution in both the training and testing datasets was heavily imbalanced, with 17.3% of samples belonging to the positive class. Such skewed distributions can hinder the model’s ability to learn generalizable features and misclassify the minority class wrongly as the majority class.[21] To combat this, two strategies were applied, oversampling the minority class and adjusting class weights in the loss function. The best performing model implemented a combination of these two strategies. Even though these measures appeared to improve results, due to time constraints, the study did not fully test which combination of class weights and oversampling would yield the best results.

This study distinguished itself from similar studies using CNN to predict RP by using a different type of model input. Similar studies typically used CT scans from a single time point. [10, 9] By using a single time point, information about temporal changes in the lung tissue occurring due to RT and inflammatory response would not be present. Therefore the model in this study used paired CT scans from before and after RT allowing the model to learn from temporal changes in the lung tissue. This approach was inspired by the study of Li et al.[11] who showed promising result by using different time points to predict disease severity in both retinopathy of prematurity and knee osteoarthritis. It is hypothesized that temporal changes captures additional information which would improve the predictive performance for RP.

The model architecture takes inspiration from ChangeNet[13], a model designed to identify differences between paired images by extracting feature maps from three semantical layers for each input. Similar to ChangeNet, features were extracted using ResNet-50 with pretrained weights. This facilitates building upon generalizable features learned from large-scale image datasets. The feature maps from both time points were concate-

nated prior to the classification. This approach allows the model to capture and learn temporal changes in lung tissue which would improve the predictive performance of RP onset over using a single CT scan alone. Although this design choice offers several benefits. The same baseline scan is potentially used multiple times, once for each follow-up scan which introduces a risk of overfitting by repeatedly exposing the model to the same data.

A major distinction from ChangeNet is how the extracted features were processed. ChangeNet is optimized for semantic segmentation, producing pixel-wise change maps to highlight structural differences.[13] In contrast, this model performs binary classification to determine whether RP develops within 90 days after a given scan. As a result, direct performance comparisons of the results between this model and ChangeNet are potentially misleading.

These fundamental differences in input and general architecture make it challenging to compare this model’s architecture directly to architectures used in other studies. Due to the lack of comparable studies using temporally paired CT scans for RP prediction, interpreting the performance of this model remains challenging.

As a part of the pre-processing, the U-net(R231) model by Hofmanninger et al.[16] was used to segment the lungs to ensure that the CNN focused on relevant lung tissue. A qualitative assessment of 40 randomly selected CT scans after lung segmentation was performed. The lung segmentation appeared successful for all inspected lungs. However, a ground truth, would have made the assessment more robust. In a study by Lee et al.[9] the U-net(r231) was also used for segmentation of the lungs and it was found that some regions of interest were removed as a result of the segmentation. Even though the lung segmentation in this study was deemed to be successful, it can not be excluded that the same problem occurred. This could mean that lung tissue containing features, strongly predictive of RP were removed, leaving the model to learn from irrelevant or non-existent patterns resulting in suboptimal performance.

The baseline and follow-up scans were acquired using different protocols for patient positioning and inspiration lev-

els, leading to misalignment and varying lung sizes due to inconsistent inspiration levels. Since the model relies on comparing anatomical changes over time, accurate alignment between scans are critical. To resolve this issue, rigid alignment was performed using coordinates for T4, T8, and T10 vertebrae. Alignment for 20 patients was manually inspected and found to be highly effective. Despite the apparent effectiveness, one follow-up scan showed unwanted rotation, which was likely due to inaccurate vertebra segmentations. Since alignment relied on the center of mass of T4, T8, and T10, poor segmentation could shift these coordinates slightly. Combined with the fact that the landmarks were all located in the spine, and are therefore relatively aligned in the xy-plane this method is susceptible to unwanted rotations. To mitigate this, one of the alignment points could be selected to be farther from the spine in the xy-plane, making the method less sensitive to this issue.

Despite rigid spine alignment, differences in lung inflation between time points may still cause lung structures to misalign. A possible solution to this problem was to implement non-rigid alignment. However it was not implemented due to concerns that such transformations might distort image features relevant for RP prediction.

To assess whether model WRR50WL can reliably predict the onset of RP within 90 days of a given scan, further research is needed. A key aspect requiring further investigation is the network architecture. It remains unclear whether the current results stem from architectural design choices or from factors such as issues with pre-processing and hyperparameter tuning. This uncertainty emphasizes the need for further research to systematically evaluate each component of our pipeline individually.

This uncertainty extends to other parts of the implementation, for instance, the limited dataset, which may have constrained the models performance. A limited dataset increases the risk of overfitting, as the model may learn patterns that do not generalize beyond the training data. With more data, it is expected that the model could better capture underlying patterns and improve generalization, ultimately leading to more robust and accurate predic-

tions.

Additionally the decision to concatenate baseline and follow-up scans should also be investigated further. An alternative approach worth exploring is the use of a difference map, created by subtracting baseline feature maps from follow-up feature maps. A difference feature map may be advantageous when comparing baseline and follow-up scans, as it more explicitly emphasizes the changes between scans while also reducing the risk of overfitting from repeated exposure to the same baseline scans.

Conclusion

The proposed model did not reliably predict the onset of RP within 90 days of a given CT scan in its current implementation, but the results still offer valuable insights and a solid foundation for future work. Many of the challenges

faced, such as architectural design uncertainties, limited dataset size and how some pre-processing methodologies were implemented can be addressed with targeted improvements. With more time and research put into these key elements it is feasible that a model like WRR50WL could yield a strong predictive performance. The findings highlight the potential of deep learning in this domain and underscores a clear direction for bettering RP predictions in future research.

Acknowledgments

UCloud interactive HPC system was used for this project, which is managed by the eScience Center at the University of Southern Denmark. Additionally, Thanks to the clinicians at Aalborg University Hospital, Department of Oncology.

References

- [1] Mandeep Singh Rahi, Jay Parekh, Prachi Pednekar, Gaurav Parmar, Soniya Abraham, Samar Nasir, Rajamurugan Subramaniyam, Gini Priyadharshini Jeyashanmugaraja, and Kulothungan Gunasekaran. Radiation-induced lung injury—current perspectives and management. *Clinics and practice*, 11(3):410–429, 2021. doi: 10.3390/clinpract11030056.
- [2] F Siddiqui, S Vaqar, and AH Siddiqui. *Lung Cancer*, 2023. URL <https://www.ncbi.nlm.nih.gov/books/NBK482357/>. Last visited: 23/05-2025.
- [3] Marek Konkol, Paweł Śniatała, and Piotr Milecki. Radiation-induced lung injury—what do we know in the era of modern radiotherapy? *Reports of Practical Oncology and radiotherapy*, 27(3):552–565, 2022. ISSN 1507–1367. doi: 10.5603/RPOR.a2022.0046.
- [4] Varsha Jain and Abigail T. Berman. Radiation pneumonitis: Old problem, new tricks. *Cancers*, 10, 7 2018. ISSN 2072-6694. doi: 10.3390/cancers10070222.
- [5] Rades D, Werner EM, Glatzel E, Bohnet S, Schild SE, Tvilsted SS, and Janssen S. Early identification of pneumonitis in patients irradiated for lung cancer—final results of the paraluc trial. *Cancers (Basel)*, 15, 2023. doi: 10.3390/cancers15020326.
- [6] Marisol Arroyo-Hernández, Federico Maldonado, Francisco Lozano-Ruiz, Wendy Muñoz-Montaño, Mónica Nuñez-Baez, and Arrieta Oscar. Radiation-induced lung injury: current evidence. *Radiology*, 314 (2):e241613, 2025. doi: 10.1186/s12890-020-01376-4.
- [7] Weronika Maria Szejniuk, Martin Skovmos Nielsen, Zsuzsanna Takács-Szabó, Jacek Pawlowski, Sahar Sulaiman Al-Saadi, Panagiotis Maidas, Martin Bøgsted, Tine McCulloch, Jens Brøndum Frøkjær, Ursula Gerda Falkmer, et al. High-dose thoracic radiation therapy for non-small cell lung cancer: a novel grading scale of radiation-induced lung injury for symptomatic radiation pneumonitis. *Radiation Oncology*, 16:1–11, 2021. doi: 10.1186/s13014-021-01857-8.
- [8] Fabian Isensee, Paul F Jaeger, Simon AA Kohl, Jens Petersen, and Klaus H Maier-Hein. nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature methods*, 18(2):203–211, 2021. doi: 10.1038/s41592-020-01008-z.
- [9] Jang Hyung Lee, Min Kyu Kang, Jongmoo Park, Seoung-Jun Lee, Jae-Chul Kim, and Shin-Hyung Park. Deep-learning model prediction of radiation pneumonitis using pretreatment chest computed tomography and clinical factors. *Technology in Cancer Research & Treatment*, 23, 2024. doi: 10.1177/15330338241254060. PMID: 38752262.
- [10] Daisuke Kawahara, Nobuki Imano, Riku Nishioka, and Yasushi Nagata. Image masking using convolutional networks improves performance classification of radiation pneumonitis for non-small cell lung cancer. *Physical and Engineering Sciences in Medicine*, 46:767–772, 2023. doi: 10.1007/s13246-023-01249-0.
- [11] Matthew D Li, Ken Chang, Ben Bearce, Connie Y Chang, Ambrose J Huang, J Peter Campbell, James M Brown, Praveer Singh, Katharina V Hoebel, Deniz Erdoğan, et al. Siamese neural networks for continuous disease severity evaluation and change detection in medical imaging. *NPJ digital medicine*, 3 (1):48, 2020. doi: 10.1038/s41746-020-0255-1.
- [12] Rodrigo Caye Daudt, Bertr Le Saux, and Alexandre Boulch. Fully convolutional siamese networks for change detection. In *2018 25th IEEE international conference on image processing (ICIP)*, pages 4063–4067. IEEE, 2018. ISBN 978-1-4799-7061-2. doi: 10.1109/ICIP.2018.8451652.
- [13] Ashley Varghese, Jayavardhana Gubbi, Akshaya Ramaswamy, and P Balamuralidhar. Changenet: A deep learning architecture for visual change detection. In *Proceedings of the European conference on computer vision (ECCV) workshops*, pages 0–0, 2018. ISBN

978-3-030-11011-6. doi: 10.1007/978-3-030-11012-3__10.

- [14] Adelisa Sirco, Ali Almisreb, Nooritawati Md Tahir, and Jamil Bakri. Liver tumour segmentation based on resnet technique. In *2022 IEEE 12th International Conference on Control System, Computing and Engineering (ICCSCE)*, pages 203–208. IEEE, 2022. doi: 10.1109/ICCSCE54767.2022.9935636.
- [15] Shiyi Wang and Guang Yang. A novel automated classification and segmentation for covid-19 using 3d ct scans. In *2022 IEEE 5th International Conference on Image Processing Applications and Systems (IPAS)*, pages 1–5. IEEE, 2022. doi: 10.1109/IPAS55744.2022.10052819.
- [16] Johannes Hofmanninger, Forian Prayer, Jeanny Pan, Sebastian Röhrich, Helmut Prosch, and Georg Langs. Automatic lung segmentation in routine imaging is primarily a data diversity problem, not a methodology problem. *European radiology experimental*, 4: 1–13, 2020. doi: 10.1186/s41747-020-00173-2.
- [17] Tugba Akinci D’Antonoli, Lucas K Berger, Ashraya K Indrakanti, Nathan Vishwanathan, Jakob Weiss, Matthias Jung, Zeynep Berkarda, Alexander Rau, Marco Reisert, Thomas Küstner, et al. Totalsegmentator mri: Robust sequence-independent segmentation of multiple anatomic structures in mri. *Radiology*, 314(2):e241613, 2025. doi: 10.1148/radiol.241613.
- [18] Sihong Chen, Kai Ma, and Yefeng Zheng. Med3d: Transfer learning for 3d medical image analysis. *arXiv preprint arXiv:1904.00625*, 2019. doi: 10.48550/arXiv.1904.00625.
- [19] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014. doi: 10.48550/arXiv.1409.1556.
- [20] Ochin Sharma. Deep challenges associated with deep learning. In *2019 international conference on machine learning, big data, cloud and parallel computing (COMITCon)*, pages 72–75. IEEE, 2019. doi: 10.1109/COMITCon.2019.8862453.
- [21] Shoujin Wang, Wei Liu, Jia Wu, Longbing Cao, Qinxue Meng, and Paul J Kennedy. Training deep neural networks on imbalanced data sets. In *2016 international joint conference on neural networks (IJCNN)*, pages 4368–4374. IEEE, 2016. doi: 10.1109/IJCNN.2016.7727770.